Health Predictor Model

Meeting note 2026-11-06

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Project Overview

The Health Predictor Model project is a student-led exercise in building a cardiovascular disease prediction system. Members will work through the machine learning process from dataset selection to model evaluation and deployment. The finished system is intended as a learning project, not a clinical diagnostic tool.

Scope and Project Goals

The project covers:

- dataset selection,
- data cleaning
- exploratory data analysis (EDA)
- Feature engineering
- model training
- model interpretation
- API
- Documentation
- Final presentation.

This will not be (Outside of Project Scope):

- Clinical diagnosis
- Medical advice or treatment recommendations
- Handling protected health information
- large-scale cloud infrastructure

Software and Tools

- Python; Pandas and NumPy for data preparation and numerical work.
- Matplotlib and Seaborn for charts and exploratory analysis.
- Scikit-learn for baseline models and evaluation; XGBoost and LightGBM for more advanced models.
- FastAPI for the prediction API; Docker for a reproducible deployment.
- GitHub for source control, issues, pull requests, and task tracking.

Team Roles and Expected Work

Members will discuss their interests and experience before roles are assigned. Roles describe the main area of responsibility; members may help across workstreams as needed.

Machine Learning Workflow

1. Data acquisition
2. Cleaning and validation
3. Exploratory data analysis
4. Feature engineering
5. Baseline models
6. Advanced models
7. Evaluation
8. Experiment tracking
9. Deployment

Team Communication

- Use Discord as the main team communication channel.
- Share meeting notes and agreed action items after each meeting.

Updated Project Schedule

Dates (2026) Planned work	
Oct 27 – Nov 2	Onboarding
Nov 2 - Nov 6	Environment Setting, VScode, Git/GitHub
Nov 3 – Nov 16	Data Selection, Data cleaning, validation, and exploratory analysis
Nov 17 – Nov 23	Feature engineering
Nov 24 – Nov 30	Baseline model training
Dec 1 – Dec 7	Advanced models and tuning
Dec 8 – Dec 14	Model interpretation and final evaluation
Dec 15 – Dec 20	Deployment and documentation

Weekly Workload Updates

Name	Completed Work last Week (Nov. 02th ~ Nov 06th) Eg. (What have you worked on this week?)(What did you find out?)
Destiny	-
Elliot	-
Moksha	-
Hendrick	-
Shahar	-

PowerPoint Link

Health Predictor Model Slides.pptx

Tasks (To be filled out by Nov. 10th)

Name	Projected Work this Week (What will you work on this week?)
Destiny	-
Elliot	-
Moksha	-
Hendrick	-
Shahar	-

Nov. 6th Meeting

- ☑ ~~GitHub Issue Progress Review~~
- ☑ ~~Dataset Selection Expectation for next week~~
- ☑ ~~Data Cleaning & Preparations~~
- ☑ ~~Feature Engineering Planning~~
- ☐ GitHub Issue assignment planned to be distributed by [Monday, November 9th]
- ☑ ~~Go over presentation on Data cleaning~~

Final Deliverables by Nov 13th Meeting

- ☐ **Research and find 3 CVD relevant Dataset**
- ☐ **Research data cleaning methods and processes**
- ☐ **Download .csv files and store it under your branch in our GitHub's -Data Folder**
- ☐ **Be prepared to share your thoughts and take ownership on the dataset you choose (Be able to explain sources, relevance, etc)**
- ☐ **Finish weekly Issue before Meeting**